

Assignment-2

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1. Pascal Triangle

Ask to the user for n number of level for triangle

Using for-loop, variables create Pascal triangle. (try to avoid array)

Program:-

```
package assignment2;
import java.util.Scanner;

public class PascalTraingle {

    public static void main(String []args){

        Scanner sc=new Scanner(System.in);
        System.out.print("Enter the number of rows: ");
        int row= sc.nextInt();

        for(int i=0;i<row;i++){
            int val=1;
            for(int space=1;space<=row-i-1;space++)
                System.out.print(" ");
            for (int j = 0; j <= i; j++) {
                if (j == 0)
                    val = 1;
                else
                    val = val * (i - j + 1) / j;
                System.out.print(val + " ");
            }
            System.out.println();
        }
        sc.close();
    }
}
```

Output:-

```
"C:\Program Files\Java\jdk-11\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 20
Enter the number of rows: 8
    1
  1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1

Process finished with exit code 0
|
```

2. Fibonacci series

Ask to the user for n number to print

Using for-loop, variables create Fibonacci series

Program:-

```
package assignment2;
import java.util.Scanner;

public class FibonacciSeries {
    public static void main(String []args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number: ");
        int value=sc.nextInt();

        int num1=0;
        int num2=1;
        System.out.print("Fibonacci Series: "+num1+" " +num2+ "
");
        for(int i=2; i<value ;i++){
            int sum= num1+num2;
            System.out.print(" " +sum+ " ");
            num1=num2;
            num2=sum;
        }
        sc.close();
    }
}
```

Output:-

```
"C:\Program Files\Java\jdk-11\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.4\lib\idea_rt.jar"
Enter the number: 10
Fibonacci Series: 0 1 1 2 3 5 8 13 21 34
Process finished with exit code 0
```